

Zuo Tianyou

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EDUCATION

Shanghai Jiao Tong University

B.S. in Electrical and Computer Engineering, Minor in Computer Science
(GPA: 3.62/4.0)

Shanghai, China

Sep. 2023 - Present
(Rank: 51/215)

COURSES

ENGR1510J	Accelerated Introduction to Computers and Programming	(4.0/4.0)
MATH1860J	Honors Mathematics II	(4.0/4.0)
CS2810J	Data Structures and Algorithms	(4.0/4.0)
CS4770J	Introduction to Algorithms	(4.0/4.0)
DS4720J	Methods and Tools for Big Data	(4.0/4.0)
CS4821J	Introduction to Operating Systems	(4.0/4.0)

PRIZES & HONORS

First Prize, NOIP 2020

Silver, UPC 2024

WORK EXPERIENCE

Alibaba | Inference Engine Intern

Jul. 2026 - Present

Worked on operator optimization for LLM inference engines, focusing on the LA(linear attention) modules in **Qwen3-Next**, including **Gated DeltaNet (GDN)** and **fused-recurrent** operators.

Moore Threads | GPU Computing & AI Software Intern

Jan. 2026 - May. 2026

Participated in the development of muBLAS, a high-performance GPU math library. Implemented and optimized CUDA kernels for linear algebra routines including LU factorization and batched operations. Analyzed shared memory usage and reduction strategies to improve numerical stability and performance.

ECE4770J Intro. to Algorithm | Teaching Assistant

Sep. 2025 - Dec. 2025

Conducted lab sessions on algorithm design and implementation in OCaml, focusing on tail recursion optimization and continuation-passing style (CPS). Maintained and extended the Online Judge system, reinforced test cases, and debugged grading scripts.

ECE2810J Data Structures and Algorithms | Teaching Assistant

Sep. 2025 - Dec. 2025

Delivered recitation classes covering advanced data structures and complexity analysis. Designed supplementary problem sets inspired by ACM/ICPC and maintained the course OJ system.

PROJECTS

muBLAS

muBLAS is Moore Threads's in-house GPU basic linear algebra library (counterpart to NVIDIA cuBLAS). Participated in the design and development of several core matrix operation APIs, including batched LU factorization `getrfBatched`, batched linear system solving `getrsBatched`, and batched matrix inversion `getriBatched` and `matinvBatched`.

LemonDB: Multi-threaded In-Memory Relational Database

[Repo](#)

Developed a high-concurrency database from scratch using **C++20**. Engineered a high-performance pull-based thread pool where worker threads fetch tasks in batches, reducing global lock contention by 16x, and utilized local queues to minimize synchronization overhead. Enforced thread safety via table-level `std::shared_mutex` with RAII guards. Built an adaptive scheduler that seamless fallbacks to single-thread mode under light workloads to eliminate thread pool overhead, and implemented a dependency-aware topology using `std::variant` and a global priority heap for automatic transaction routing.

Mum Shell

[Repo](#)

Built a Unix-like shell in C supporting redirection, pipes, background jobs, and signal handling. Implemented process control using `fork`, `exec`, and `waitpid`.

MSD Distributed Recommendation System

[Repo](#)

Built an end-to-end big data pipeline on Million Song Dataset; compressed 288GB HDF5 data into 884MB Avro (99.7% efficiency gain) and optimized PySpark processing time from 5.5 to 1.3 hours. Implemented sub-second similarity recommendations using FAISS (HNSW/LSH) and achieved 80% accuracy within a ± 10 -year error margin using Spark MLlib and XGBoost.

SKILLS

Programming Languages	C/C++, CUDA, Python, Shell, OCaml
Language Skills	Mandarin, English
Tech Skills	Git, SSH, CUDA Kernel Development, vLLM, Linux, Docker, Spark